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Practical 09

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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CODE :-

```
ishak@YASHAWINI MSYS ~  
$ nano optimal.c
```

```
GNU nano 8.7 optimal.c  
#include <stdio.h>  
  
int main() {  
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int frames[3];  
    int n = 12, frames_count = 3;  
    int i, j, k, pos, farthest;  
    int hit = 0, fault = 0;  
    int flag;  
  
    for(i = 0; i < frames_count; i++)  
        frames[i] = -1;  
  
    for(i = 0; i < n; i++) {  
        flag = 0;  
  
        // Check for page hit  
        for(j = 0; j < frames_count; j++) {  
            if(frames[j] == pages[i]) {  
                hit++;  
                flag = 1;  
                break;  
            }  
        }  
  
        // Page fault  
        if(flag == 0) {  
            pos = -1;  
            farthest = i + 1;  
  
            for(j = 0; j < frames_count; j++) {  
                int found = 0;
```



GNU nano 8.7

op

```
    pos = -1;
    farthest = i + 1;

    for(j = 0; j < frames_count; j++) {
        int found = 0;
        for(k = i + 1; k < n; k++) {
            if(frames[j] == pages[k]) {
                if(k > farthest) {
                    farthest = k;
                    pos = j;
                }
                found = 1;
                break;
            }
        }

        if(found == 0) {
            pos = j;
            break;
        }
    }

    if(pos == -1)
        pos = 0;

    frames[pos] = pages[i];
    fault++;
}

float hit_percent = (hit * 100.0) / n;
float fault_percent = (fault * 100.0) / n;

printf("Total Page Hits: %d\n", hit);
printf("Total Page Faults: %d\n", fault);
printf("Page Hit Percentage: %.2f%%\n", hit_percent);
printf("Page Fault Percentage: %.2f%%\n", fault_percent);

return 0;
}
```

OUTPUT:-

```
ishak@YASHAWINI MSYS ~  
$ gcc optimal.c -o optimal && ./optimal  
Total Page Hits: 5  
Total Page Faults: 7  
Page Hit Percentage: 41.67%  
Page Fault Percentage: 58.33%  
  
ishak@YASHAWINI MSYS ~  
$ -
```